

LabX & CDS — Complete Cheat Sheet

Cognizant Training Reference | LabX License, User Management, Instruments, Workspaces, CDS Basics, OpenLab CDS & ECM

1. LabX — INTRODUCTION & PURPOSE

What is LabX?

LabX (by METTLER TOLEDO) is a laboratory data management and instrument control software enabling centralized control, monitoring, and documentation of results from balances, titrators, pH meters, density meters, refractometers, melting point systems, and Karl Fischer titrators.

- Automated data collection, standardized procedures, secure handling of results.
- Supports electronic records and audit trails — ideal for GxP, GLP, 21 CFR Part 11 compliance.

Purpose

- Centralized instrument control and method management
- Automatic result collection and reporting
- Seamless integration with LIMS, ERP, and other lab systems
- User management with roles, permissions, and audit trails
- Enhanced data quality, integrity, compliance, and productivity

2. SYSTEM SPECIFICATIONS & ARCHITECTURE

System Specs

- Client/Server system — every attached computer and instrument is treated as a client.
- Express Edition: up to 3 instruments on one local computer (Standalone).
- Server Edition: distributed or localized, up to 30 instruments per server.
- Clients: Windows executables — NOT web-based. Servers: hosted in Windows services.
- Database: MS SQL Server or Oracle.

Architecture Modes

- Standalone: All components (DB, app, service) on one computer.
- Distributed (Preferred): Components spread across separate computers.
- Business Server connects to at least 3 Instrument Servers; each connects 7–15 instruments.
- Instruments connect via Ethernet — no physical lab PC needed per instrument.
- LabX v11: up to 30 instruments (or 60 with Server Full Edition) per MS SQL DB instance.

Key Components

- Client-Server: UI on workstations; server manages data, processing, instrument comm.
- Communication: USB, Ethernet, Serial (industry-standard protocols).
- Security: User auth, data encryption, audit trails.
- Integration: API-based with LIMS and other systems.
- Scalability: Supports multiple users/instruments; future-ready.

3. LICENSE MANAGEMENT & ACTIVATION

License Procurement — Step by Step

Step 1: Choose Starter Pack — Express (local, ≤3 instruments) or Server (30/60 instruments, multiple computers). With Express-Server license you can Sidegrade Express to Server.

Step 2: Configure Options — User Management, Product Database, Auto Import/Export, Statistics/Trending, Regulation (21 CFR/EU Annex 11, Server only), Report Designer, System Integration (API/LIMS/ELN/ERP), STAReX Integration.

Step 3: Choose Instruments — LabX licensed per connected instrument. Supported: Analytical/Precision Balances, Titrators, Karl Fischer, pH Meters, UV/Vis, Refractometers, Density Meters, Melting Point Devices, Quantos.

Step 4: Software Care — Standard, Comprehensive, or On-Demand service.

License Summary & Activation

- Each system needs an Edition license key + one Instrument license per instrument.
- Non-activated LabX runs 45 days (Demo); cannot use after 60 days.
- Online Activation: requires internet — use <http://activation.mt.com>.
- Offline Activation: generate License Key File → transfer to internet PC → upload to Portal.
- Reactivation required within 10 days if licenses added/removed or hardware replaced.

4. USER MANAGEMENT & ROLE CONFIGURATION

Overview

Define User Roles and Permissions to control access. Located at System workspace > User Management. Supports Local users and Active Directory (AD)-linked users.

Managing Users

- Add/change/delete/temporarily disable users.
- Default local user at installation: PredefinedUser.
- Add Local User: System > User Management > User > New User > fill details > save.
- AD Users: Delegates user/password management to AD. With Role Mapping enabled, roles auto-synced on first login.

Roles

- Roles define rights available to users — each user needs at least one role; multiple roles are cumulative.
- Default: 5 roles — can be added, edited, deleted.
- Rights subdivided by system and modules.

5. SIGNATURE & RELEASE POLICIES

- Global settings — apply to ALL LabX users on all workstations.
- Objects cannot be released until all required signatures obtained.
- Types: None (immediate Approved), Review (1 signature), Review–Approve (2 signatures).
- Auto Release: when enabled, versions release automatically with last required signature.

6. WORKSPACES

Workspace	Purpose	Key Functionalities
Data and Results	Organizing experimental data; storing & viewing results	Data Visualization, Organization, Result Management, Data Integrity & Security
Analysis and Method	Conducting data analysis; developing analytical methods	Analysis Tools, Method Development, Workflow Mgmt, Compliance
Resources and Instruments	Instrument Management & Resource allocation	Instrument Connection/Control, Resource Mgmt, Scheduling
System	Audit Trail, Background Jobs, User Management	Audit Trail, Background Jobs, User Mgmt/Roles, Task Import/Export

7–8. INSTRUMENTS & COMMUNICATION SETUP

Supported METTLER TOLEDO Instruments

Balance, Density Refractometer, Seven Excellence, Thermal Value, Titrator, UV/Vis, Quantos, Peripheral Devices (Barcode reader, Fingerprint reader).

Initial Setup on Instrument

- Language Settings: Select desired language from instrument panel.
- Date/Time Setup: Choose display format; additional steps vary by instrument type.
- Network Settings: Set IP, Subnet, Gateway if using Ethernet; enable DHCP reservation if needed.

- LabX Connection: Define communication type, connection type (network/USB), port (default 8014).
- After restart, instrument panel shows 'Connecting to LabX...'

Note: Date/time on instruments synchronized with LabX Instrument Server at every connection. UI language changed to LabX installation language on connection.

9. ADDING AN INSTRUMENT

Login to LabX Client with 'Edit Resources' rights → Resources > Instruments → New Instrument. Wizard guides: select Instrument Category → Connection Type (Ethernet/Serial/USB) → Instrument Server → enter settings → LabX tests connection → enter name/alias → Finish.

Note: LabX is licensed per connected instrument — one license per instrument only. At least one free instrument license must be available.

10. MANAGING INSTRUMENTS

State	Meaning
Ready	Available, license installed, compatible, no hardware differences
Busy	Instrument is busy
Interaction Required	User interaction needed before task can continue
Blocked	Instrument is blocked
Connected	Instrument is connected to LabX
Error / System error	Error occurred on instrument
Connecting	Currently connecting to LabX
Disconnected	Instrument cannot be reached

Network Interruption: Instrument disconnects; active task paused. Data from completed steps stored in DB. On reconnection, user can resume or restart. No data loss.

11. INSTALLATION

- LabX delivered on write-protected USB flash drive.
- Standalone: All components on one computer.
- Distributed: Application components across different computers.
- Run Setup file (LabXSetup.exe) → Install .NET Framework 4.8 → reboot → select LabX System type.
- Standalone: SQL Server installed automatically. Distributed: Select Custom path → modify parameters.

12–13. DATA MANAGEMENT, ANALYSIS & WORKFLOW

Data Management

- Automatic raw data recording from instruments in proprietary format.
- Central DB with user access control, audit trails, encryption.
- Import/Export in various formats; versioning & backup; metadata management.

Data Analysis & Workflow

- Built-in: statistical calculations, curve fitting, trend analysis, data visualization.
- Customizable & exportable reports; user-defined calculations; scripting & automation.
- Compliance: 21 CFR Part 11, GLP, GMP via compliance templates and electronic signatures.
- Integration with LIMS: seamless data transfer for sample management and traceability.

14–15. DATA BACKUP & MIGRATION

Backup Options

- Automatic Scheduled: regular backups at desired intervals to specific location.
- Automatic Real-time Replication: continuous mirror copy on secondary server.
- Manual Export: entire LabX DB (.mdf/.bak) or specific projects.
- Best Practices: periodic test restores, monitor logs, use encryption, maintain Disaster Recovery Plan.

Migration

- Upgrading LabX Version / Moving to new system / Relocating DB to new server.
- Use METTLER TOLEDO Migration Tool or LabX DB Maintenance Tool (Restore DB with .bak file).
- Example v9→v11: backup .bak → transfer to new DB Server → use 'Publish Archive' → view archived data in 'View Only' mode.

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— CDS (Chromatography Data System) Reference —

16. CDS — BASICS & FUNDAMENTALS

What is CDS?

A computer-based system to control chromatographic instruments, acquire detector signals, process data, and generate results and reports. Ensures data integrity and regulatory compliance (21 CFR Part 11). Controls GC, LC (HPLC), CE, IC, GPC/SEC, GC-MS, LC-MS systems.

Evolution of CDS

Era	Characteristics
Pre-1970s — Manual Era	Paper strip-chart recorders; peaks measured manually using rulers and planimeters
1970s–1980s — Electronic Integrators	Stand-alone integrators for automated peak area and height calculations
1980s–1990s — PC-Based CDS	Computers connected to instruments for digital data acquisition, integration, and storage
Late 1990s–2000s — Client-Server CDS	Networked systems with centralized servers, user access control, audit trails, 21 CFR Part 11 compliance
2010s–Present — Enterprise & Cloud CDS	Centralized, scalable CDS integrated with LIMS/ELN; high data integrity, compliance, remote access, smart analytics

Basic Requirements of a CDS

- Manage growing volume of chromatography data quickly and efficiently.
- Expandable architecture: stand-alone, thin-client, or client/server environments.
- Maintain data adherence to GxP / 21 CFR Part 11.
- Acquire, process, report, review, and approve data in a streamlined series of operations.
- Historical data readily available at any time during retention period.
- All data modifications tracked efficiently.

Globally Adopted CDS Solutions

CDS Solution	Approx. Market Share
Empower™ (Waters)	~35%
Chromeleon™ (Thermo)	~25%
OpenLab™ (Agilent)	~16%
LabSolutions™ (Shimadzu)	~13%
Others (Compass, Clarity, TRILUTION...)	~11%

Note: Worldwide ~70–80% usage share: Empower, Chromeleon, OpenLab, LabSolutions.

17. INTRODUCTION TO CHROMATOGRAPHY

Term coined by Russian botanist Mikhail Tswett (1906). Chromatography separates mixture components by differences in affinity between a mobile phase and an immiscible stationary phase. Used in pharmaceutical, petroleum, food & beverage, cosmetic, forensic, and diagnostic industries.

Chromatographic Techniques

LC/HPLC, Gas Chromatography (GC), Ion Chromatography, Capillary Electrophoresis (CE), Thin Layer Chromatography (TLC), Paper Chromatography, Gel Permeation/Size Exclusion Chromatography (GPC/SEC).

Detectors

LC: UV-Vis, Photo Diode Array (PDA), Refractive Index* (RI), Fluorescence, Charged Aerosol (CAD), Conductivity, Mass Spectroscopy*.

GC: Flame Ionization (FID), Electron Capture (ECD), Thermal Conductivity* (TCD), Flame Photometric (FPD), Mass Spectroscopy*.
(* = Universal Detectors)

18. BASICS OF HPLC & GC INSTRUMENTS

HPLC Components

Solvent Reservoir → Degasser → Pump → Injector → Column (in Column Oven) → Detector → Recorder/Data System → Waste.

Typical Systems: Waters Alliance, Agilent HPLC/UPLC, Shimadzu HPLC, Waters UPLC.

GC Modules

- Carrier gas and other gases (per detector type).
- Samplers: Injector and Headspace.
- Inlet(s), Column and Column Oven.
- Detector (FID, TCD, ECD, etc.).
- Data Acquisition & Control System (e.g., OpenLab CDS, Empower CDS, Chromeleon CDS).

Typical GC Systems: Agilent 7693 Series ALS, Perkin Elmer, 7697A Headspace Sampler.

19. AGILENT OPENLAB CDS — OVERVIEW

Agilent OpenLab CDS is an integrated software platform designed to acquire, control, process, and manage chromatographic data across laboratories (~16% global market share).

Key Highlights

- Centralized data system for LC, GC, LC/MS, and GC/MS.
- Supports Agilent and multi-vendor instruments — Thermo, Shimadzu, Perkin, Metrohm, Waters.
- Combines instrument control, data processing, and reporting.
- Designed for regulated and non-regulated environments.
- Enables secure, compliant, and efficient laboratory operations.
- Compliant with 21 CFR Part 11, GxP, GLP regulatory requirements.

Key Features & Business Value

Category	Details
Core Features	Centralized data management with secure access control; Regulatory compliance: 21 CFR Part 11, audit trails, e-signatures; Scalable: Standalone, Client/Server, or Cloud; Intuitive UI; Integration with OpenLab ECM, LIMS, ELN
Business Value	Improved data integrity and compliance; Increased lab productivity and standardization; Reduced IT complexity and operational risk; Supports digital transformation of QC and R&D; labs

*Note: *21 CFR Part 11 refers to Title 21 of the U.S. Code of Federal Regulations, Part 11 — Electronic Records & Signatures.*

OpenLab CDS Product Comparison

Feature	OpenLab Client/Server	OpenLab CDS (2.x)	OpenLab ChemStation
Category	IT Architecture / Topology	Software Application (New)	Software Application (Legacy)
Primary Goal	Centralization & Networking	Data Integrity & Ease of Use	Flexibility & Familiarity
User Interface	N/A (Depends on App)	Modern Ribbon-style	Classic Windows-style
Instrument Control	Managed by AICs	Integrated (LC/GC/MS)	Integrated (Highly Flexible)
Latest Version	v2.8 (Server/ECM XT)	v2.8	LTS 01.11
OS Support	Windows Server 2022 / Cloud	Windows 10/11 / AWS / Azure	Windows 11 (LTS only)

Feature	OpenLab Client/Server	OpenLab CDS (2.x)	OpenLab ChemStation
Compliance	Highest (Centralized)	Built-in by design	Add-on (requires OpenLab ECM)

Business Integration Scenario

OpenLab CDS bridges LIMS and Chromatograph instruments: LIMS provides sample information → CDS handles instrument control, sample set, data acquisition, data processing, and data storage → Chromatograph instruments execute runs.

Note: Goal of CDS: assure regulatory compliance, protect intellectual property, minimize risk of data loss, support costs, and maximize business values.

20. OPENLAB ECM — SDMS OVERVIEW

Laboratory instrumentation produces large amounts of data (machine and human readable) used throughout drug discovery and development. This data represents significant intellectual property. OpenLab ECM (from Agilent Technologies) is a Knowledge Engineering System designed to store, organize, and protect critical data — including binary data files and human readable instrument output.

Why SDMS is Needed — Data must be:

- Collected, warehoused, and maintained in a central storage location accessible to scientists for review.
- Secure and handled in a manner that meets regulatory and patent guidelines.
- Traceable from upload to archive through a single source.
- Accessible for restoration so review and analysis can be performed within the original environment.

Data Transfer Methods to OpenLab ECM

- Electronic data files can be "printed" to an electronic file and transferred via ECM Print Services.
- Files can be manually uploaded/downloaded by the user.
- Files can be uploaded automatically by an OpenLab Scheduler Agent.

21. OPENLAB SYSTEM ARCHITECTURE

OpenLab ECM has an n-tier architecture allowing modular functionality. The topology consists of an Oracle database server, an Application Server, a Web server, File Transfer servers, a file storage device, and an archived file storage device. It can also include one or more Scheduler Agent devices.

Architecture Components

Component	Function
Web Server	Hosts the application component; configured under IIS using http/https based on user requirement.
File Transfer Server	Used to transfer files to primary storage.
Application Server	Used to configure application components.
Scheduler Agent Server	Used to set up tasks for file sweep.
File Storage	Primary storage location to keep physical files on a permanent basis.
Database	Used to keep application information in database.

22. OPENLAB ADMINISTRATION MODULE

Account Administration

- Configuration — system-wide settings.
- Users/Group — manage user accounts and groups.
- Storage — manage storage configurations.
- Electronic Signature — configure e-signature settings.
- Keys — manage encryption/access keys.
- User Preferences — individual user settings.

Global Administration

- Configure Email — set up email notifications.
- Create Account — create new system accounts.

- License Manager — manage product licenses.
- System Errors — view and manage system error logs.
- System Status — monitor overall system health.

23. USER ACCESS MANAGEMENT (OpenLab ECM)

Three-tier model for access control:

Entity	Purpose
Users	Used to create individual user accounts with personal credentials.
Groups	Used to create user groups to organize users with common roles/permissions.
Roles	Used to define roles and associate them with user groups to grant specific access rights.

Default Roles in OpenLab ECM

- Approver
- Archive Administrator
- Archive Agent
- BPMS User
- Contributor
- Folder Administrator
- Reader
- Retention Manager
- Scheduler Administrator
- System Administrator
- ESignatures

Note: Roles can be reset to defaults using the 'Reset Roles' button in the Roles administration panel.

24. ECM CLIENT INSTALLATION

OpenLab ECM client works with Google Chrome and IE browser. However, Agilent recommends ECM Classic client features to access the application from a local system. End Users need to install ECM Client on the local system in order to access the classic client.

Installer Files to be installed on user's machine:

- AGTBHELRL Reranme.msi
- AGT-HA (File).msi
- AGT301 Client Chrome Installation.msi
- AGTWS-V1 Client-Client Installation Update Tool.msi
- AGTWS-VCI Client Installation Update Server.msi
- AGT4U-TITU Client Uninstallation Update.msi
- AGT v2.01 Application Setup Installer.msp
- FBAETCV Installation Agilent Installation.msi
- AGTMS Administration (Application).msi
- WhatsIm (ie (web)).exe
- LL-MMI Preancitrate.txt (XML Document)

Note: Exact file names/versions may vary by deployment. Consult your system administrator for the correct installer package.

25. COA, ELECTRONIC RECORDS & SIGNATURES

Certificate of Analysis (COA)

For every sample, a Certificate of Analysis (COA) is required before any product can be released to the open market for sale or usage. It lists the parameters, details of tests performed, and methods used. A COA verifies that the product has been tested against defined parameters and complies with regulatory and quality standards.

21 CFR Part 11 — Electronic Records & Signatures

Requirement	Rule
Electronic Records	Must be accurate, complete, protected, and retrievable at all times.
Electronic Signatures	Legally binding if they meet FDA requirements under 21 CFR Part 11.

ECM vs CDS — Core Functional Difference

System	Primary Role
OpenLab ECM (SDMS)	Data archival platform — ensures secure long-term storage and compliance with regulatory requirements.
OpenLab CDS	Data acquisition platform — captures and manages chromatographic data for analysis and reporting.

26. CDS COMPARISON: EMPOWER vs CHROMELEON vs OPENLAB

Feature / Criteria	Waters Empower	Thermo Chromeleon	Agilent OpenLab CDS
Vendor Ecosystem Strength	Very strong with Waters instruments; limited openness	Strong across Thermo + good multi-vendor	Strongest openness for multi-vendor (Agilent + 3rd party)
Multi-Vendor Support	Moderate	Good	Excellent
Architecture	Traditional client-server; database-centric (Oracle)	Client-server + domain-based architecture	Client-server + evolving toward cloud (OpenLab XT)
Cloud Readiness	Limited (mostly on-prem)	Limited / partial	Strong roadmap (OpenLab XT cloud-native)
Scalability (Enterprise)	Proven for large labs but complex to scale	Good scalability	Very strong, especially with XT backend
Ease of Deployment	Moderate (DB-heavy setup)	Easier than Empower	Moderate but improving with standardization
Data Management	Central DB; tight but rigid	Flexible data vaults	Highly flexible + modern data handling
User Interface	Older UI, less intuitive	Modern, intuitive UI	Modern UI (especially newer versions)
Method Development	Strong chromatographic capabilities	Excellent (especially IC workflows)	Strong, especially in HPLC/GC workflows
Instrument Control	Best for Waters instruments	Excellent for Thermo	Broadest support including legacy systems
Compliance (21 CFR Part 11)	Industry gold standard	Strong	Strong
Audit Trails & Data Integrity	Very robust, mature	Robust	Robust, more flexible
Report Customization	Good but sometimes rigid	Very flexible	Very flexible
Integration (LIMS/ELN)	Possible but can be complex	Good integration options	Strong enterprise integration (APIs, connectors)
Customization Capability	Limited	Moderate	High
Performance (Large Data)	Can slow due to DB dependency	Good performance	Strong, especially with XT architecture
Remote Access / Web	Limited	Partial	Improving; stronger in XT/cloud model
Upgrade Complexity	Complex (DB upgrades painful)	Easier than Empower	Moderate
Total Cost of Ownership	High (infra + DB + support)	Moderate	Moderate to high (better ROI in large setups)
Best Fit Use Case	Regulated labs (QC, pharma)	Labs using Thermo ecosystem	Large enterprises, multi-site, cloud adoption

27. CDS PLATFORM — STRENGTHS & LIMITATIONS

Platform	Best For	Limitations
Waters Empower	Highly regulated QC labs; organizations heavily invested in Waters; proven compliance with minimal audit risk.	Not cloud-friendly; less flexible for multi-vendor environments; higher maintenance overhead.
Thermo Chromeleon	Balanced usability + compliance; labs with Ion Chromatography (IC) focus; mixed vendor setups at moderate scale.	Not as future-ready for cloud as OpenLab; slightly less enterprise modularity.
Agilent OpenLab CDS	Enterprise-scale multi-instrument environments; digital transformation / cloud roadmap; integration-heavy ecosystems (LIMS, MES, ELN).	Initial implementation complexity; requires architecture planning (especially XT/cloud).

28. CDS DECISION MATRIX

Scenario / Environment	Recommended CDS
Strict regulatory QC lab, stable on-prem infrastructure	Waters Empower
Balanced usability + compliance, moderate scale, Thermo instruments	Thermo Chromeleon
Large enterprise, cloud roadmap, multi-vendor, integration-heavy	Agilent OpenLab CDS
Ion Chromatography (IC) focused workflows	Thermo Chromeleon
Multi-site deployment with centralized data governance	Agilent OpenLab CDS (XT)

Summary: Empower = stability & compliance (traditional stronghold). Chromeleon = usability + balanced capabilities. OpenLab = future-ready, scalable, enterprise platform.